Contents

1 Some Data-Cleaning Tools **1**
1.1 Rectangular and non-rectangular data . . . 1
1.2 Tidy Data . . . 1
1.3 Tools for tidying data . . . 2

1 Some Data-Cleaning Tools

- reading: @WickhamTidyData2014

1.1 Rectangular and non-rectangular data

Figure 1: Tweet by Nathan Oseroff-Spicer (nathanoseroff): 'It makes sense, though, since there are rows and columns. It's scientific'. Source: <https://twitter.com/nathanoseroff/status/1321565407568867332>

- Data represented in rows and columns is called *rectangular data*.
- Common non-rectangular forms include XML and JSON
  - Sometimes you might get data in the form of a PDF or image. This is called *cursèd data*.

1.2 Tidy Data

- conventions for organizing data for analysis
  1. Each variable forms a column.
  2. Each observation forms a row.
  3. Each type of observational unit forms a table.
- Example: student demographics and major, for students who took a philosophy course at UC Davis, 2005-2015

- slightly rephrased
  - 1 table = 1 unit of observation
    - student-term (major)
    - student (demographics)
  - 1 row = 1 unique observed unit
    - 1 row = 1 student-term combination
    - 1 row = 1 student
  - Every column is either an *identifier* or a *measured variable*
    - rule of thumb: 1 row = 1 unique combination of identifiers
    - major table
      - identifiers: student ID, `term`
      - measured variables: admit term, relative term, major
      - NB the `term` identifier is constructed from two measured variables
    - demographics table

id	admit_term	rel_term	term	major
000[redacted]B68	201010	YR1FALL	201010	LPOL-Political Science
000[redacted]B68	201010	YR1WINTER	201101	LPOL-Political Science
000[redacted]B68	201010	YR1SPRING	201103	LPOL-Political Science
000[redacted]B68	201010	YR1SUMMER_I	201105	LPOL-Political Science
000[redacted]B68	201010	YR1SUMMER_II	201107	LPOL-Political Science
000[redacted]B68	201010	YR2FALL	201110	LPOL-Political Science
000[redacted]B68	201010	YR2WINTER	201201	LPOL-Political Science

id	gender	ethnicity_code	ethnicity	race	first_gen	low_income	admission_type	ever_phil	poc	race4	demographic	major_at_first_phil
8F[redacted]73	F	OT	Other	Other	FALSE	FALSE	Transfer	FALSE	FALSE	Other	F.Other	LENL-English
0F[redacted]99C	M	(Missing)	Other	Other	TRUE	FALSE	Transfer	FALSE	FALSE	Other	M.Other	APME-Pre-Managerial Economics
1C[redacted]9C2	F	WH	White/Caucasian	White	TRUE	FALSE	Transfer	FALSE	FALSE	White	F.White	LPSC-Psychology
3E[redacted]BEA	F	MX	Mexican-Am/Mexican/Chicano	Hispanic	TRUE	FALSE	Transfer	TRUE	TRUE	BHIP	F.BHIP	LPHI-Philosophy
95[redacted]693	F	KO	Korean-American/Korean	Asian	TRUE	TRUE	Transfer	FALSE	TRUE	Asian	F.Asian	AHDE-Human Development
A1[redacted]6B0	M	CH	Chinese-American/Chinese	Asian	FALSE	TRUE	Freshman	FALSE	TRUE	Asian	M.Asian	EMEC-Mechanical Engineering
6C[redacted]127	M	FP	Filipino/Filipino-American	Asian	FALSE	FALSE	Transfer	FALSE	TRUE	Asian	M.Asian	ABIS-Biological Sciences

Figure 2: Two tables in a tidy dataset. The first table records students' majors during each term they were enrolled; the second (next figure) records demographics. The tables have different units of observation, but can be linked using the ID (internal to this dataset) for an analysis at the level of individual students. Other tables in this dataset include final grades for philosophy courses taken by the students, and demographics of the instructors of those courses.

- identifier: student ID
- measured variables: gender, race and ethnicity variables, first gen status, low income status, admission type, whether they ever majored in philosophy, major the semester they took their first philosophy course
- NB here "measured variable" includes both variables in the raw data as well as variables I constructed for analysis purposes (`poc`, `demographic`)

1.3 Tools for tidying data

```r
library(tidyverse)
```

```
## -- Attaching packages --------------------------------------- tidyverse 1.3.0 --

## v ggplot2 3.3.2     v purrr   0.3.4
## v tibble  3.0.4     v dplyr   1.0.2
## v tidyr   1.1.2     v stringr 1.4.0
## v readr   1.4.0     v forcats 0.5.0

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
```

```r
theme_set(theme_minimal())

data_folder = file.path('data')
data_zip = file.path(data_folder, 'tidy.zip')
## Get the data
if (!file.exists(data_zip)) {
    download.file('https://github.com/data-science-methods/book/raw/master/data/tidy.zip',
                  data_zip)
}
unzip(data_zip, overwrite = FALSE, exdir = data_folder)
```

```
## Warning in unzip(data_zip, overwrite = FALSE, exdir = data_folder): not
```

```
## overwriting file 'data/billboard.csv
```

```
## Warning in unzip(data_zip, overwrite = FALSE, exdir = data_folder): not
## overwriting file 'data/weather.csv
```

```
## Warning in unzip(data_zip, overwrite = FALSE, exdir = data_folder): not
## overwriting file 'data/tb.sav
```

```
## Warning in unzip(data_zip, overwrite = FALSE, exdir = data_folder): not
## overwriting file 'data/pew.csv
```

1.3.1 3.1: The Pew Dataset

- pivot_longer()
- if_else()
- fct_inorder()

```
pew_df = read_csv(file.path(data_folder, 'pew.csv'))
```

```
## Warning: Missing column names filled in: 'X1' [1]
```

```
##
## -- Column specification --------------------------------------------------------
## cols(
##   X1 = col_double(),
##   religion = col_character(),
##   `<$10k` = col_double(),
##   `$10-20k` = col_double(),
##   `$20-30k` = col_double(),
##   `$30-40k` = col_double(),
##   `$40-50k` = col_double(),
##   `$50-75k` = col_double(),
##   `$75-100k` = col_double(),
##   `$100-150k` = col_double(),
##   `>150k` = col_double(),
##   `Don't know/refused` = col_double()
## )
```

```
head(pew_df)
```

X1	religion	<$10k	$10-20k	$20-30k	$30-40k	$40-50k	$50-75k	$75-100k	$100-150k	>150k
1	Agnostic	27	34	60	81	76	137	122	109	84
2	Atheist	12	27	37	52	35	70	73	59	74
3	Buddhist	27	21	30	34	33	58	62	39	53
4	Catholic	418	617	732	670	638	1116	949	792	633
5	Don't know/refused	15	14	15	11	10	35	21	17	18
6	Evangelical Prot	575	869	1064	982	881	1486	949	723	414

```
pew_clean = pew_df %>%
    ## Drop the row number column
    select(-X1) %>%
    ## Lengthen: 1 row per religion x income level
    pivot_longer(cols = `<$10k`:`Don't know/refused`,
                 names_to = 'income',
                 values_to = 'freq') %>%
    ## Cleaning `income`:
    ## - Make "Don't know/refused" into NA
    ## - Coerce to ordered factor
```

```r
    mutate(income = if_else(str_detect(income, "Don't know"),
                            NA_character_,
                            income),
           income = fct_inorder(income))
head(pew_clean)
```

religion	income	freq
Agnostic	<$10k	27
Agnostic	$10-20k	34
Agnostic	$20-30k	60
Agnostic	$30-40k	81
Agnostic	$40-50k	76
Agnostic	$50-75k	137

```r
ggplot(pew_clean, aes(religion, freq, fill = fct_rev(income))) +
    geom_col() +
    scale_fill_viridis_d(na.value = 'grey50') +
    coord_flip()
```

```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <80>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
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## <99>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <e2>

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## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <80>

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## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
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## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <e2>
```

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## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
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## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
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## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
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## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
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## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
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## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <80>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <99>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
```

```
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
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## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
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## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
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## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <99>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <80>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <99>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <80>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <99>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
```

```
## <80>
```

```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <99>
```

```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <e2>
```

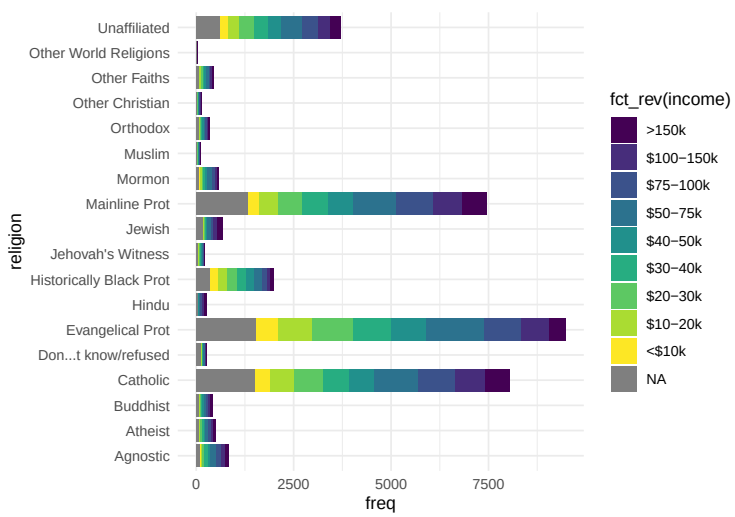
```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <80>
```

```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <99>
```

```
## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <e2>
```

```
## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <80>
```

```
## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <99>
```



1.3.2 3.1: The Billboard Dataset

- A little helper function for parsing
- `str_remove()`
- Coercion
- Date arithmetic

```
billboard_df = read_csv(file.path(data_folder, 'billboard.csv'))
```

```
## Warning: Missing column names filled in: 'X1' [1]
```

```
##
```

```
## -- Column specification --------------------------------------------------------
## cols(
##   .default = col_double(),
##   artist = col_character(),
##   track = col_character(),
##   time = col_time(format = ""),
##   date.entered = col_date(format = ""),
##   wk66 = col_logical(),
##   wk67 = col_logical(),
##   wk68 = col_logical(),
##   wk69 = col_logical(),
##   wk70 = col_logical(),
##   wk71 = col_logical(),
##   wk72 = col_logical(),
##   wk73 = col_logical(),
##   wk74 = col_logical(),
##   wk75 = col_logical(),
##   wk76 = col_logical()
## )
## i Use `spec()` for the full column specifications.
```

```r
head(billboard_df)
```

X1	year	artist	track	time	date.entered	wk1	wk2	wk3	wk4	wk5	wk6
1	2000	2 Pac	Baby Don't Cry	04:22:00	2000-02-26	87	82	72	77	87	94
2	2000	2Ge+her	The Hardest Part Of ...	03:15:00	2000-09-02	91	87	92	NA	NA	NA
3	2000	3 Doors Down	Kryptonite	03:53:00	2000-04-08	81	70	68	67	66	57
4	2000	3 Doors Down	Loser	04:24:00	2000-10-21	76	76	72	69	67	65
5	2000	504 Boyz	Wobble Wobble	03:35:00	2000-04-15	57	34	25	17	17	31
6	2000	98^0	Give Me Just One Nig...	03:24:00	2000-08-19	51	39	34	26	26	19

```r
#' Helper function to parse weeks to integers in the Billboard dataset
#' @param wk_chr Vector of week characters, with the pattern `c('wk1', 'wk2')`
#' @return Vector of integers, eg, `c(1L, 2L)`
parse_week = function(wk_chr) {
    wk_int = wk_chr %>%
        str_remove('wk') %>%
        as.integer()
    return(wk_int)
}

billboard_clean = billboard_df %>%
    ## Drop the row number column
    select(-X1) %>%
    ## Pivot week down and remove NA rows
    pivot_longer(cols = starts_with('wk'),
                 names_to = 'week',
                 values_to = 'rank') %>%
    filter(!is.na(rank)) %>%
    ## Parse week and calculate date
    ## NB Base R can handle date arithmetic
    ## For month arithmetic see ?lubridate::`%m+%`
    mutate(week = parse_week(week),
           date = date.entered + (week-1)*7) %>%
    ## Clean columns
```

```
    select(year, artist, time, track, date, week, rank)
head(billboard_clean)
```

year	artist	time	track	date	week	rank
2000	2 Pac	04:22:00	Baby Don't Cry	2000-02-26	1	87
2000	2 Pac	04:22:00	Baby Don't Cry	2000-03-04	2	82
2000	2 Pac	04:22:00	Baby Don't Cry	2000-03-11	3	72
2000	2 Pac	04:22:00	Baby Don't Cry	2000-03-18	4	77
2000	2 Pac	04:22:00	Baby Don't Cry	2000-03-25	5	87
2000	2 Pac	04:22:00	Baby Don't Cry	2000-04-01	6	94

NB. Base R includes methods for doing date arithmetic, eg,

```
as.Date('2020-04-30') + 1
```

```
## [1] "2020-05-01"
```

If you need month arithmetic, check out the `lubridate` package, and specifically `%m+%` and related operators.

```
count(billboard_clean, artist, track) %>%
    arrange(desc(n)) %>%
    head()
```

artist	track	n
Creed	Higher	57
Lonestar	Amazed	55
3 Doors Down	Kryptonite	53
Hill, Faith	Breathe	53
Creed	With Arms Wide Open	47
Joe	I Wanna Know	44

1.3.3 3.2: The TB Dataset

- `haven` package
- `separate()`
- `case_when()`

```
tb_df = haven::read_sav(file.path(data_folder, 'tb.sav'))
head(tb_df)
```

country	year	m014	m1524	m2534	m3544	m4554	m5564	m65	mu	f014	f1524	f2534	f3544	f4554
AD	2000	0	0	1	0	0	0	0	NA	NA	NA	NA	NA	NA
AE	2000	2	4	4	6	5	12	10	NA	3	16	1	3	0
AF	2000	52	228	183	149	129	94	80	NA	93	414	565	339	205
AG	2000	0	0	0	0	0	0	1	NA	1	1	1	0	0
AL	2000	2	19	21	14	24	19	16	NA	3	11	10	8	8
AM	2000	2	152	130	131	63	26	21	NA	1	24	27	24	8

```
tb_clean = tb_df %>%
    ## One row per observation = country x gender x age group
    pivot_longer(m014:fu,
                 names_to = 'column',
                 values_to = 'cases') %>%
    ## Separate gender and age
    separate(column, into = c('gender', 'age'),
             sep = 1) %>%
    ## Wrangle age:
    ## - Split into upper and lower bounds
```

```r
    ## - Recombine with -
    ## - Fix missing age and 65+
    ## - Set order
    ## - Clean up the intermediate columns for upper and lower bounds
    separate(age, into = c('age_lower', 'age_upper'),
             sep = -2) %>%
    mutate(age = str_c(age_lower, '-', age_upper),
           age = case_when(age == '-u'  ~ NA_character_,
                           age == '-65' ~ '65+',
                           TRUE         ~ age),
           age = fct_inorder(age)) %>%
    select(-age_lower, -age_upper) %>%
    ## Put in order, with ID columns first and then the variable
    select(country, year, gender, age, cases)
head(tb_clean)
```

country	year	gender	age	cases
AD	2000	m	0-14	0
AD	2000	m	15-24	0
AD	2000	m	25-34	1
AD	2000	m	35-44	0
AD	2000	m	45-54	0
AD	2000	m	55-64	0

1.3.4 3.3: The Weather Dataset

- `pivot_wider()`
- `str_remove()`

```r
weather_df = read_csv(file.path(data_folder, 'weather.csv'))
```

```
## Warning: Missing column names filled in: 'X1' [1]
```

```
##
## -- Column specification --------------------------------------------------------
## cols(
##   .default = col_double(),
##   id = col_character(),
##   element = col_character(),
##   d9 = col_logical(),
##   d12 = col_logical(),
##   d18 = col_logical(),
##   d19 = col_logical(),
##   d20 = col_logical(),
##   d21 = col_logical(),
##   d22 = col_logical(),
##   d24 = col_logical()
## )
## i Use `spec()` for the full column specifications.
```

```r
head(weather_df)
```

X1	id	year	month	element	d1	d2	d3	d4	d5	d6	d7	d8	d9	d10	d11	d12
1	MX17004	2010	1	tmax	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
2	MX17004	2010	1	tmin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
3	MX17004	2010	2	tmax	NA	27.3	24.1	NA	NA	NA	NA	NA	NA	NA	29.7	NA
4	MX17004	2010	2	tmin	NA	14.4	14.4	NA	NA	NA	NA	NA	NA	NA	13.4	NA
5	MX17004	2010	3	tmax	NA	NA	NA	NA	32.1	NA	NA	NA	NA	34.5	NA	NA
6	MX17004	2010	3	tmin	NA	NA	NA	NA	14.2	NA	NA	NA	NA	16.8	NA	NA

```r
weather_clean = weather_df %>%
    select(-X1) %>%
    ## Lengthen days
    pivot_longer(d1:d31,
                 names_to = 'day',
                 values_to = 'obs') %>%
    ## Widen `element`
    pivot_wider(names_from = element,
                values_from = obs) %>%
    ## Fix the class of `day`
    mutate(day = str_remove(day, 'd'),
           day = as.integer(day)) %>%
    ## Check against the paper by filtering to non-missing rows
    filter(!is.na(tmax) | !is.na(tmin))
head(weather_clean)
```

id	year	month	day	tmax	tmin
MX17004	2010	1	30	27.8	14.5
MX17004	2010	2	2	27.3	14.4
MX17004	2010	2	3	24.1	14.4
MX17004	2010	2	11	29.7	13.4
MX17004	2010	2	23	29.9	10.7
MX17004	2010	3	5	32.1	14.2